Hrishikesh Pawar

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Education

Worcester Polytechnic Institute **August 2023 - May 2025**
Master of Science in Robotics Engineering; **GPA: 4.0/4.0** *Worcester, MA*
Courses: **CS541**-Deep Learning, **RBE549**-Computer Vision, **RBE595**-Vision based Robot Manipulation, **REB550**-Motion Planning
Teaching Assistant: **RBE595**-Vision based Robot Manipulation, **RBEBL01**-Robots for Recycling

Experience

PeAR Lab | *Graduate Researcher advised by Prof. Nitin Sanket* | *Worcester, MA* **August 2024 – Present**
- Developed a Depth-from-Defocus framework with event cameras for computationally efficient autonomous navigation.
- Implementing RL policy for adaptive control of optical parameters, improving navigation efficiency and obstacle avoidance.
- Developed a custom **3D Gaussian Splats** simulator for **high frame-rate**, realistic frames and events, enabling **HITL** testing.

Nobi | *AI Software Intern* | *Remote* **May 2024 - August 2024**
- Led R&D efforts for smart ceiling lamps improving real-time fall detection and emergency response in elderly care.
- Engineered a **rotation-aware** detection model by integrating **Swin Transformers,** improving accuracy from **85 mAP** to **92 mAP**.
- Experimented with **LLaVa** and **CLIP**, applying **LoRA** for **PEFT** fine-tuning for vision-language detection task generalization.
- Reduced deployment time by **30%**, automating end-to-end deployment pipeline using **Jenkins**, **Kubernetes** and **Docker**.

Adagrad AI | *Computer Vision Engineer* | *Pune, India* **November 2020 - July 2023**
- Worked on R&D for **Gate-Guard**, an edge-based Boom Barrier system using **Automatic License Plate Recognition (ALPR)**.
- Achieved accuracy of **97%** for four-wheelers and **95%** for two-wheelers with **50 FPS** throughput on **Nvidia Jetson-TX2.**
- Designed interactive analytics and monitoring services using **Django**, **Azure**, **WebSockets**, **Kafka**, **Celery** and **Redis**.
- Featured links: Indian Express | Times of India | YouTube

Publications

Hrishikesh Pawar*, Deepak Singh*, Nitin J. Sanket, **"Blurring for Clarity: Passive Computation for Defocus-Driven Parsimonious Navigation using a Monocular Event Camera"**, Accepted for presentation at **EVGEN Workshop**, **IEEE WACV, March 2025**.
- Introduced a biologically inspired approach leveraging optical defocus in **event cameras** for **parsimonious** aerial navigation.
- Applied optical defocus to extract **sharpness-based depth cues**, segmenting foreground obstacles from the background.
- Achieved **70%** success rate with **62x** computational savings compared to state-of-the-art methods.

Projects

Deep Visual-Inertial Odometry | GitHub | Project Report
- Developed a VIO stack, coupling **CNNs** for image processing and **LSTMs** for sequential inertial data enhancing pose estimation.
- Generated synthetic datasets in **Blender** simulating trajectories with realistic sensor noise to improve model robustness.
- Trained Visual-Inertial models combining **MSE** and **Geodesic Loss** reducing the RMSE **Absolute Trajectory Error** by **28%**.
- Won the **best project** award for the **Computer Vision course** (**RBE-549**) at **WPI** amongst **40** graduate students.

Zero-Shot Semantic Neural Style Transfer for Images | GitHub
- Implemented AdaAttN module enabling per-point attentive normalization for **zero-shot semantic neural style transfer**.
- Achieved a **13%** reduction in training time without loss of style transfer quality by removing deeper attention layers.
- Integrated **AdaAttN** with **CLIPSeg's** prompt-based segmentation, enabling user-defined, localized neural style transfer.

Skills

- **Languages**: Python, C, C++, SQL
- **Frameworks**: PyTorch, OpenCV, TensorFlow, ONNX, NumPy, Pandas, Django, Flask, Celery, Kafka, MATLAB, CMake, Docker, TensorRT, Jenkins, Kubernetes, Git, ROS, ROS2